

JPQL, Native SQL and Projections with a Student Grade Tracker

The academic team needs more than raw grade data — they need aggregated insights. The current API returns every grade record for every student in one flat response with no way to derive averages, find top scorers, or get a clean summary view. All the basic endpoints are already working. Your job is to progressively improve the API by introducing custom JPQL queries, a native SQL query, and projections so the academic team gets exactly the insights they need.

What's already provided:

- `Student.java` — entity with `id`, `name`, `email`
- `Grade.java` — entity with `id`, `subject`, `score`, `@ManyToOne` to `Student`
- `StudentRepository.java` — `JpaRepository`, no custom methods
- `GradeRepository.java` — `JpaRepository`, no custom methods yet
- `GradeService.java` — `getAllGrades()` returns a flat `List<Grade>`
- `GradeController.java` — `GET /api/grades` returns all grades at once
- `DataLoader.java` — seeds 5 students with 4 subject grades each on startup
- `application.properties` — H2 in-memory database, ready to use

Specifically, you are expected to:

1. Hit the existing endpoint and experience the problem firsthand — all 20 grade records returned in one flat response with no aggregation or structure.
2. Add a custom `@Query` using JPQL to calculate the average score for a given student name and wire it to a new endpoint.
3. Add another JPQL query to find the highest scoring grade record for a given subject and wire it to a new endpoint.
4. Add a native SQL query using `nativeQuery = true` that returns the average score for every student in one query — understand when and why you reach for native SQL over JPQL.
5. Define a projection interface and use it to return only student name and average score instead of full entity objects.

Deliverables:

- GET /api/grades returns all grade records.
- GET /api/grades/average?student=Alice returns Alice's average score.
- GET /api/grades/top?subject=Math returns the top scoring grade for that subject.
- GET /api/grades/averages returns average scores for all students via native SQL.
- GET /api/grades/summary returns a projected view with only name and average — no full entity.

Testing Endpoints

Experience the problem first:

```
curl -s http://localhost:8080/api/grades | cat
```

Average score for a student:

```
curl -s "http://localhost:8080/api/grades/average?student=Alice" | cat
curl -s "http://localhost:8080/api/grades/average?student=Bob" | cat
curl -s "http://localhost:8080/api/grades/average?student=Carol" | cat
```

Top scorer for a subject:

```
curl -s "http://localhost:8080/api/grades/top?subject=Math" | cat
curl -s "http://localhost:8080/api/grades/top?subject=Science" | cat
curl -s "http://localhost:8080/api/grades/top?subject=History" | cat
```

All student averages via native SQL:

```
curl -s http://localhost:8080/api/grades/averages | cat
```

Summary projection — name and average only:

```
curl -s http://localhost:8080/api/grades/summary | cat
```